

CHIP 2026 Hackathon — Team J5 (ITB)

Phase 1 Report: NTT/INTT Baseline (C0)

Every Phase 1 artifact: RTL, verification (lint, simulation on two simulators, formal), Quartus implementation on Cyclone V 5CSEBA6U23I7, and the timing-failure root-cause analysis

Status: **PARTIAL** · Branch phase1-ntt-baseline · HEAD d76c829 (working tree has uncommitted Quartus evidence) · not pushed

Timing NOT met: worst setup slack -48.323 ns at the provisional 20.000 ns clock; Fmax 14.64 MHz. C0 is a baseline measurement, not an optimisation result.

Approval box in docs/results/result_phase1.md: NOT ticked.

Written for: Team J5, as the review record before deciding how Phase 1 closes. The repository files are the source of truth; this PDF is generated from them (docs/evidence/phase01-ntt-baseline/, docs/evidence/quartus/, docs/results/result_phase1.md) plus fresh lint/slang/pytest runs.

1. Summary

Configuration C0 is the Phase 1 baseline: one butterfly per cycle ($L = 1$), an unbanked 256x12 polynomial memory, a generated twiddle ROM, and plain % modular reduction. It is functionally correct (bit-exact against the golden model on Icarus and Verilator, constant cycle count, formally proven FSM safety property) and it has now been implemented by Quartus. The Quartus result is the first MEASURED hardware-implementation data point of the project: it fits comfortably but misses timing badly.

Gate	Criterion	Status
CRG-1	Lint clean (verilator -Wall)	PASS
CRG-2	Elaboration clean (slang)	PASS
CRG-3	Bit-exact vs golden, Icarus and Verilator	PASS
CRG-4	Corner cases listed before tests	PASS
CRG-5	Phase 0 regression still passes	PASS
CRG-6	Locked parameters	PASS
CRG-7	Constant cycle count	PASS
CRG-8	Formal properties (safety)	PASS
CRG-9	Quartus evidence; no negative slack or failure documented	FAIL
CRG-10	Result artifact + claim checker	PASS
1b-1	Target-clock ADR recorded	MISSING
1b-2	C0 ablation row filled with MEASURED values	PASS

CRG-9 is recorded as FAIL. The roadmap wording ("or the failure documented") could be read as PASS once documented; that reading is a team decision, not taken here.

Headline numbers (MEASURED; denominators as printed by the fitter)

Quantity	Value
Logic utilization (ALMs)	7,010 / 41,910 (17 %)
Registers	3104
RAM blocks (M10K)	0 / 553 (0 %)
Block memory bits	0 / 5,662,720 (0 %)
DSP blocks	3 / 112 (3 %)
Fmax (Slow 1100mV 100C / -40C)	14.64 MHz / 14.69 MHz
Worst setup slack / TNS (@ 20.000 ns)	-48.323 ns / -143688.194 ns
Worst hold slack	0.211 ns (Fast 1100mV -40C) - met
NTT / INTT cycles (simulation)	897 / 1153, constant over all tested inputs

2. Analysis of the Quartus C0 evidence (the nine questions)

Q1. Did every stage complete?

Stage	Quartus message (verbatim)
Analysis & Synthesis	Quartus Prime Analysis & Synthesis was successful. 0 errors, 8 warnings
Fitter	Quartus Prime Fitter was successful. 0 errors, 2 warnings
Assembler	Quartus Prime Assembler was successful. 0 errors, 1 warning
Timing Analyzer	Quartus Prime Timing Analyzer was successful. 0 errors, 5 warnings

Yes - all four completed with 0 errors. Completion is not timing closure: the Timing Analyzer raised *Critical Warning* (332148): *Timing requirements not met* 4 times, once per corner.

Q2. Measured values

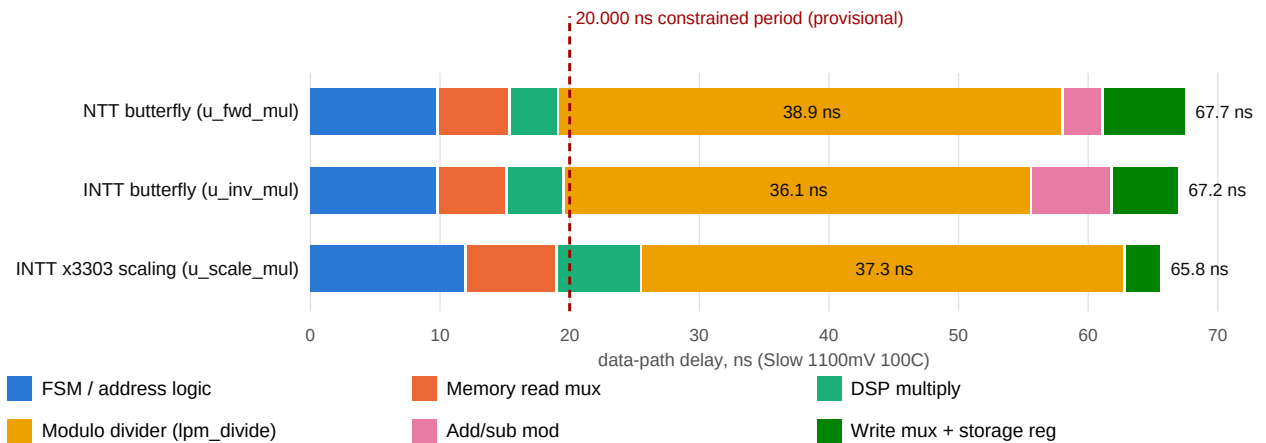
Corner / check	Slack (ns)	TNS (ns)
Slow 1100mV 100C Setup	-48.323	-143688.194
Slow 1100mV 100C Hold	0.531	0.000
Slow 1100mV 100C Minimum Pulse Width	9.255	0.000
Slow 1100mV -40C Setup	-48.059	-143216.393
Slow 1100mV -40C Hold	0.562	0.000
Slow 1100mV -40C Minimum Pulse Width	9.149	0.000
Fast 1100mV 100C Setup	-17.039	-49249.473
Fast 1100mV 100C Hold	0.230	0.000
Fast 1100mV 100C Minimum Pulse Width	9.566	0.000
Fast 1100mV -40C Setup	-11.893	-34032.465
Fast 1100mV -40C Hold	0.211	0.000
Fast 1100mV -40C Minimum Pulse Width	9.557	0.000

Source: C0.sta.summary (docs/evidence/quartus/C0-20260929.md). Cross-check (derived): $1 / (20.000 + 48.323) \text{ ns} = 14.64 \text{ MHz}$, equal to the Fmax panel.

Q3. Why timing fails (from the reports)

The worst path starts at `layer_q[2]` and ends at `poly_mem:u_mem|mem[64][5]`. Its data delay is 67.684 ns against a 20.000 ns period (arrival 73.704 ns, required 25.381 ns). The path is the whole butterfly step done in one cycle: the FSM's per-layer length lookup and address arithmetic, the read multiplexer of the register-based memory, the DSP multiply, a **combinational 24-bit / 12-bit divider** that Quartus inferred from % (Info 278004, `lpm_divide`, 146 cells on this path), the add/sub modulo, and the write multiplexer back into a memory register. Every one of the 3072 memory registers ends such a path.

Q4. Which structure limits the clock?



Path	FSM / address logic	Memory read mux	DSP multiply	Modulo divider (lpm_divide)	Add/sub mod	Write mux + storage reg	Total	Slack
NTT butterfly (u_fwd_mul)	9.952	5.504	3.780	38.926	3.060	6.462	67.684	-48.323
INTT butterfly (u_inv_mul)	9.952	5.294	4.381	36.096	6.194	5.234	67.151	-47.787
INTT x3303 scaling (u_scale_mul)	12.068	7.010	6.540	37.329	0.000	2.846	65.793	-46.444

Delay per structure in ns, derived by quartus/phase01_ntt_c0/segment_path.py from the Quartus `report\_timing -detail full\_path` output: each row's incremental delay (cell or the routing into it) is charged to the structure named in the element path. Rules are in the script.

Primary limiter: modular reduction. The lpm_divide is 38.926 ns (57.5 %) of the worst path, 36.096 ns on the INTT path and 37.329 ns on the scaling path; by itself it is longer than the 20 ns period. **Not the only limiter:** on the same worst path the remaining segments (address logic 9.952, memory read mux 5.504, DSP 3.780, add/sub 3.060, write mux 6.462 ns) sum to 28.758 ns, also more than 20 ns - so the single-cycle read-compute-write structure of the butterfly datapath and the register-based memory contribute too. **Control logic:** a control-only path (layer_q[1] to layer_q[2]) fails by 3.829 ns and is 92.7 % interconnect, with one 11.986 ns hop from row Y37 to row Y4; the zeta-index path passes with 0.351 ns. **Clock/constraint method:** not the cause (Q6).

Q5. Worst path, start point and end point

Item	Value
Startpoint	layer_q[2]
Endpoint	poly_mem:u_mem mem[64][5]
Launch / latch clock	clk_i / clk_i (20.000 ns relationship)
Launch clock path	6.020 ns
Latch clock path	5.441 ns (pessimism removed 2.185 ns)
Clock uncertainty	-0.060 ns
Data delay	67.684 ns = cells 29.959 + interconnect 37.725
Slack	-48.323 (VIOLATED)

Cells on the worst path (cumulative ns from the launch register, increment, element). Carry-chain cells inside the divider are omitted; each divider stage (op_N sumout, StageOut) is kept.

```

0.000 +0.000 layer_q[2]|q
1.166 +0.538 Mux7~0|combout
2.037 +0.308 ShiftRight0~3|combout
2.809 +0.544 ShiftRight0~5|combout
3.920 +0.480 ShiftLeft0~0|combout
5.322 +0.912 Add0~13|sumout
6.584 +0.932 Add2~5|sumout
8.019 +1.169 Add3~1|cout
8.357 +0.338 Add3~25|sumout
9.952 +0.354 mem_addr_b[3]~1|combout
12.091 +0.079 u_mem|Mux15~47|combout
13.895 +0.222 u_mem|Mux15~51|combout
14.615 +0.508 u_mem|Mux15~62|combout
15.456 +0.512 u_mem|Mux15~84|combout

```

```

19.236 +3.100 u_bfly|u_fwd_mul|Mult0-mac|resulta[13]
22.125 +0.359 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_4~1|sumout
23.058 +0.097 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[145]~108|combout
25.040 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_5~1|sumout
25.909 +0.084 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[157]~93|combout
27.991 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_6~1|sumout
29.484 +0.084 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[170]~26|combout
31.420 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_7~1|sumout
32.386 +0.097 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[183]~21|combout
34.035 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_8~1|sumout
35.172 +0.087 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[198]~27|combout
37.114 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_9~1|sumout
38.064 +0.084 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[213]~95|combout
40.063 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_10~1|sumout
41.035 +0.081 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[224]~10|combout
43.023 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_11~1|sumout
44.035 +0.083 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[237]~33|combout
45.794 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_12~1|sumout
47.206 +0.303 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[248]~6|combout
49.371 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_14~1|sumout
50.471 +0.083 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[261]~4|combout
52.224 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_15~1|sumout
53.321 +0.079 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[280]~12|combout
55.017 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_16~1|sumout
55.803 +0.564 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[293]~35|combout
57.326 +0.154 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|op_17~5|sumout
58.162 +0.080 u_bfly|u_fwd_mul|Mod0|auto_generated|divider|divider|StageOut[300]~0|combout
59.361 +0.301 u_bfly|Add6~9|shareout
59.867 +0.506 u_bfly|Add6~13|cout
59.867 +0.000 u_bfly|Add6~17|cout
59.917 +0.050 u_bfly|Add6~21|cout
59.917 +0.000 u_bfly|Add6~25|cout
59.967 +0.050 u_bfly|Add6~33|cout
59.967 +0.000 u_bfly|Add6~37|cout
60.031 +0.064 u_bfly|Add6~49|cout
60.031 +0.000 u_bfly|Add6~29|cout
60.447 +0.416 u_bfly|Add6~41|sumout
61.222 +0.558 u_bfly|LessThan2~2|combout
61.968 +0.498 mem_wdata_b[0]~0|combout
63.014 +0.080 mem_wdata_b[5]~11|combout
64.946 +0.518 mem_wdata_b[5]~5|combout
67.422 +0.075 u_mem|mem[64][5]~feeder|combout
67.684 +0.262 poly_mem:u_mem|mem[64][5]

```

All 3104 setup endpoints

Endpoint type	Endpoints				Violating	
poly_mem storage register	3072				3072	
scale_addr_q	8				0	
p_q	8				0	
zeta_idx_q	7				0	
state_q	4				0	
layer_q	3				2	
mode_q	1				0	
done_o	1				0	
Slack range (ns)	[-50, -40)	[-40, -30)	[-30, -20)	[-20, -10)	[-10, 0)	[0, 25)
Endpoints	3072	0	0	0	2	30

Every violating endpoint's worst path starts at layer_q[*]. Sum of the negative per-endpoint slacks = -143688.194 ns, equal to the STA TNS.

Q6. Does the virtual-pin clock change the conclusion?

Critical Warning (15725): clock port is fed by virtual pin "clk_i-input"; timing analysis treats input to the clock port as a ripple clock

Info (15717): Design contains 39 virtual pins; timing numbers associated with paths containing virtual pins are estimates

It affects accuracy, not the conclusion. Because clk_i is a virtual pin, the clock enters from a logic cell (LABCELL_X1_Y36_N15) and Quartus calls it a ripple clock; it still reaches the global clock network (clk_i~CLKENA0 on CLKCTRL_G3, fan-out 3104). Launch and latch use the same network, so on the worst path the clock contributes skew -0.579 ns and uncertainty 0.060 ns - about 0.64 ns of the 48.323 ns violation - while the data delay alone is 67.684 ns. Absolute clock insertion delays (about 5-6 ns here) are not representative of a real clock pin or PLL, and the

23 input / 14 output ports are unconstrained, so I/O timing is not analysed; the failing paths are all register-to-register.

Q7. Were the supplied reports enough? What else was generated

No. C0.sta.rpt contains only summary panels (Fmax, slack per corner, unconstrained ports); it has no path detail, so it cannot name the critical path. The missing report is a detailed `report_timing`. It was generated read-only from the existing post-fit netlist (no recompile, RTL/QSF/SDC unchanged) with `quartus_sta -t quartus/phase01_ntt_c0/report_critical_paths.tcl`, which runs `report_timing -setup -npaths 10 -detail full_path`, a 5000-path per-endpoint summary, per-instance -through reports for `u_fwd_mul / u_inv_mul / u_scale_mul`, control-register reports and the hold report. GUI equivalent: Tools > Timing Analyzer > Reports > Custom Reports > Report Timing... (Setup, `clk_i`, detail Full path, 10+ paths), or Tasks > Macros > Report Top Failing Paths.

Q8 / Q9. What was deliberately not done

The SDC was not loosened and the target clock was not lowered to make timing pass. No RTL was modified. The critical path is identified and documented first; any change is a separate, reviewed step in the phase the roadmap assigns it to (arithmetic: Phase 5; pipelining: Phase 4; memory: Phase 2), unless the team decides to change C0 itself.

3. Quartus implementation detail

What synthesis built

Info (276004): RAM logic "twiddle_rom:u_rom|rom_zeta" is uninferred due to inappropriate RAM size File: /home/ajil/FPGA/Projects/CHIPATON/rtl/ntt/twiddle_rom.sv Line: 19
Info (278004): Inferred divider/modulo megafunction ("lpm_divide") from the following logic: "butterfly:u_bfly|modmul_reduce:u_inv_mul|Mod0" File: /home/ajil/FPGA/Projects/CHIPATON/rtl/ntt/modmul_reduce.sv Line: 26
Info (278004): Inferred divider/modulo megafunction ("lpm_divide") from the following logic: "modmul_reduce:u_scale_mul|Mod0" File: /home/ajil/FPGA/Projects/CHIPATON/rtl/ntt/modmul_reduce.sv Line: 26
Info (278004): Inferred divider/modulo megafunction ("lpm_divide") from the following logic: "butterfly:u_bfly|modmul_reduce:u_fwd_mul|Mod0" File: /home/ajil/FPGA/Projects/CHIPATON/rtl/ntt/modmul_reduce.sv Line: 26
Info (12134): Parameter "LPM_WIDTHN" = "24"
Info (12134): Parameter "LPM_WIDTHD" = "12"

Hierarchy node	ALMs needed (own)	Comb. ALUTs (own)	Registers (own)
ntt_core	7010.0 (155.3)	4144 (201)	3104 (32)
butterfly:u_bfly	463.8 (76.6)	907 (143)	0 (0)
modmul_reduce:u_fwd_mul	196.0 (0.0)	388 (0)	0 (0)
modmul_reduce:u_inv_mul	191.2 (0.0)	376 (0)	0 (0)
modmul_reduce:u_scale_mul	189.9 (0.0)	376 (0)	0 (0)
poly_mem:u_mem	6171.0 (6171.0)	2624 (2624)	3072 (3072)
twiddle_rom:u_rom	29.7 (29.7)	36 (36)	0 (0)

From the Fitter 'Resource Utilization by Entity' panel. Each modmul_reduce's ALMs sit entirely in its lpm_divide child. poly_mem is 6171 of the 7010 ALMs needed and all 3072 memory registers: the 256x12 array with two asynchronous read ports was built from flip-flops and multiplexers, not M10K.

```
; butterfly:u_bfly|modmul_reduce:u_inv_mul|Mult0~mac ; Two Independent 18x18 ; DSP_X20_Y4_N0 ;
; modmul_reduce:u_scale_mul|Mult0~mac ; Two Independent 18x18 ; DSP_X86_Y2_N0 ;
; butterfly:u_bfly|modmul_reduce:u_fwd_mul|Mult0~mac ; Two Independent 18x18 ; DSP_X20_Y6_N0 ;
```

Three DSP blocks in 'Two Independent 18x18' mode, unsigned, input/output registers not used.

Constraint file used (quartus/phase01_ntt_c0/C0.sdc)

```
create_clock -name clk_i -period 20.000 [get_ports {clk_i}]
derive_clock_uncertainty

set_false_path -from [get_ports {rst_ni}]
```

4. Verification artifacts (simulation, lint, formal)

RTL, cocotb tests and formal files are unchanged since commit d8cb3cd (git diff --stat d8cb3cd -- rtl/tb/ntt/ formal/ is empty), so the simulation and formal logs below apply to the same RTL that Quartus compiled. Lint, slang, pytest and check_params were re-run fresh for this report.

Test plan (CRG-4, written before tests) -

docs/evidence/phase01-ntt-baseline/test_plan.md

```
# Phase 1 test plan – NTT/INTT baseline (C0), written before any test is coded (CRG-4)

Scope: `rtl/ntt/modmul_reduce.sv`, `rtl/ntt/base_case_multiply.sv`, `rtl/ntt/butterfly.sv`,
`rtl/ntt/twiddle_rom.sv`, `rtl/ntt/poly_mem.sv`, `rtl/ntt/ntt_core.sv`. Reference: `tb/golden/primitives.py`
(`ntt`, `intt`, `base_case_multiply`). No RTL below was written before this plan.

## Unit: modmul_reduce (a*b mod q)

- Exhaustive is  $3329^2 \approx 11.08\text{M}$  pairs – too slow for per-commit cocotb, so Phase 1 uses randomized coverage; the exhaustive sweep required by Phase 5 (`docs/ROADMAP.md` §5) is deferred there.
- Corner cases: a=0, b=0; a=q-1, b=q-1 (max product); a=1, b=anything (identity); a=q-1, b=1.
- 2000 random pairs a,b  $\in [0,q)$ .

## Unit: base_case_multiply (Algorithm 12)

- Corner cases: all-zero inputs; a0=a1=q-1, b0=b1=q-1; gamma=0; gamma=q-1; gamma = an actual  $\zeta^{(2\text{-BitRev7}(i)+1)}$  value taken from the golden model (i=0 and i=127, the two ends of the table).
- 500 random (a0,a1,b0,b1,gamma) tuples, gamma restricted to real values from the golden model's `_GAMMA` table (never an arbitrary value, since real hardware only ever uses those 128 constants).

## Unit: butterfly (CT forward / GS inverse)

- Corner cases: a=0,b=0; a=q-1,b=q-1; zeta=1 (i=0 entry, unused in practice but must not corrupt math if ever driven); zeta = the largest tabulated value; a=0,b=q-1 and a=q-1,b=0 (boundary of the subtraction-then-add-q path).
- 1000 random (a,b,zeta) tuples per mode (forward, inverse), zeta restricted to the golden model's `_ZETA_BITREV` table.

## Unit: ntt_core (top-level FSM, both directions)

- Corner cases (fed as the full 256-coefficient input polynomial):
  1. All-zero polynomial.
  2. All coefficients = q-1 (max value).
  3. Impulse: f[0]=1, rest 0.
  4. Impulse at the last coefficient: f[255]=1, rest 0.
  5. Alternating 0 / q-1.
  6. A polynomial whose NTT is itself an all-zero or all-(q-1) pattern is not assumed; only literal input patterns are used as corner cases, per the rule against inventing vectors.
- Property tests, cross-checked against `tb/golden/primitives.py` bit-exact:
  - `ntt(f)` matches `primitives.ntt(f)` for every corner case and 100 random polynomials.
  - `intt(f_hat)` matches `primitives.intt(f_hat)` for every corner case and 100 random polynomials.
  - `intt(ntt(f)) == f` for 50 random polynomials (round-trip through the RTL itself, not just each direction separately).
- **Constant-cycle check (CRG-7):** run `ntt_core` on every corner case and 20 random polynomials in the *same direction*; the number of clock cycles from `start_i` to `done_o` must be identical for all of them (the algorithm has no data-dependent branch, so this must hold by construction; the test makes that an explicit, logged fact rather than an assumption).
- Run twice: once under Verilator, once under Icarus Verilog (CRG-3); a mismatch between the two is investigated before trusting either result.

## Formal (CRG-8, SymbiYosys)

- FSM reaches `done_o` from any legal reset state within a bounded number of cycles (`cover`), and never deasserts `busy_o` without having asserted `done_o` for exactly one cycle first.
- All addresses driven into `poly_mem` (`j`, `j+len` for the butterfly pass; the scan address for the INTT scaling pass) stay in `[0, 255]` for every reachable state (`assert`).

## Explicitly out of scope for Phase 1 (see docs/ROADMAP.md "Not allowed yet")

Memory banking or multiple lanes, pipelining beyond one butterfly per cycle, Montgomery/Barrett reduction, lazy reduction, Karatsuba, Keccak, HPS integration, any performance/speed-up claim.
```

Lint - fresh run (verilator --lint-only -Wall)

```
- V e r i l a t i o n   R e p o r t: Verilator 5.053 devel rev v5.052-183-g39e2d8e85 (mod)
- Verilator: Built from 0.175 MB sources in 9 modules, into 0.070 MB in 3 C++ files needing 0.000 MB
- Verilator: Walltime 0.061 s (elab=0.013, cvt=0.028, bld=0.000); cpu 0.028 s on 1 threads; allocated 20.305 MB
exit=0
```

Elaboration - fresh run (slang)

Top level design units:
ntt_core

Build succeeded: 0 errors, 0 warnings
exit=0

cocotb regression, both simulators -

docs/evidence/phase01-ntt-baseline/cocotb_regression_2026-09-29.txt

Phase 1 (C0) cocotb regression -- both simulators, 2026-09-29 03:50:09 UTC
=====

```
--- Icarus Verilog ---
[icarus] modmul_reduce (test_modmul): 2/2 PASS
[icarus] base_case_multiply (test_base_case_multiply): 2/2 PASS
[icarus] butterfly (test_butterfly): 2/2 PASS
[icarus] ntt_core (test_ntt_core): 4/4 PASS
[icarus] TOTAL: 10/10 passed

--- Verilator ---
[verilator] modmul_reduce (test_modmul): 2/2 PASS
[verilator] base_case_multiply (test_base_case_multiply): 2/2 PASS
[verilator] butterfly (test_butterfly): 2/2 PASS
[verilator] ntt_core (test_ntt_core): 4/4 PASS
[verilator] TOTAL: 10/10 passed
```

CRG-7 constant-cycle count (ntt_core, C0, from test_constant_cycle_count, identical on both simulators): NTT = 897 cycles, INTT = 1153 cycles (896/1152 butterfly-pass cycles + 1 cycle of start/done overhead), measured across the 5 corner-case polynomials (docs/evidence/phase01-ntt-baseline/test_plan.md) plus 20 random polynomials per direction, in each direction separately. Single value observed in every case -- no variance.

Test counts per docs/evidence/phase01-ntt-baseline/test_plan.md:

modmul_reduce:	5 corner cases + 2000 random pairs
base_case_multiply:	6 corner cases + 500 random tuples
butterfly:	6 corner cases x2 modes + 1000 random x2 modes
ntt_core (NTT):	5 corners + 100 random, bit-exact vs tb/golden/primitives.py:ntt
ntt_core (INTT):	5 corners + 100 random, bit-exact vs tb/golden/primitives.py:intt
ntt_core (roundtrip):	50 random, intt(ntt(f)) == f through the RTL
ntt_core (CRG-7):	5 corners + 20 random, both directions

No test was weakened, no vector invented; all reference values come from tb/golden/primitives.py.

Formal safety proof (SymbiYosys) -

docs/evidence/phase01-ntt-baseline/formal_ntt_core_safety_2026-09-29.txt

```
SBY 10:57:10 [ntt_core_safety] engine_0.induction: ## 0:00:00 Trying induction in step 3..
SBY 10:57:10 [ntt_core_safety] engine_0.induction: ## 0:00:00 Temporal induction successful.
SBY 10:57:10 [ntt_core_safety] engine_0.induction: ## 0:00:00 Status: passed
SBY 10:57:10 [ntt_core_safety] engine_0.induction: finished (returncode=0)
SBY 10:57:10 [ntt_core_safety] engine_0.induction: Status returned by engine for induction: pass
SBY 10:57:10 [ntt_core_safety] engine_0.basecase: ## 0:00:00 Status: passed
SBY 10:57:10 [ntt_core_safety] engine_0.basecase: Status returned by engine for basecase: pass
SBY 10:57:10 [ntt_core_safety] summary: Elapsed clock time [H:MM:SS (secs)]: 0:00:00 (0)
SBY 10:57:10 [ntt_core_safety] summary: Elapsed process time [H:MM:SS (secs)]: 0:00:00 (0)
SBY 10:57:10 [ntt_core_safety] summary: engine_0 (smtbmc boolector) returned pass for basecase
SBY 10:57:10 [ntt_core_safety] summary: engine_0 (smtbmc boolector) returned pass for induction
SBY 10:57:10 [ntt_core_safety] summary: engine_0 did not produce any traces
SBY 10:57:10 [ntt_core_safety] summary: successful proof by k-induction.
SBY 10:57:10 [ntt_core_safety] DONE (PASS, rc=0)
Property: exactly one cycle after busy_o drops, done_o is 1. Liveness is not formally proven (evidence from simulation). Address range [0,255] is argued structurally (8-bit addresses).
```

Phase 0 regression and locked parameters - fresh runs

```
..... [100%]
23 passed in 22.06s

ok      Q      = 3329 (locked 3329; verified: IETF draft-cfrg-schwabe-kyber-03; FIPS 203)
ok      N      = 256 (locked 256; verified: same)
ok      ZETA   = 17 (locked 17; verified: primitive 256th root of unity mod q (tcgcrest notes))
ok      K      = 3 (locked 3; FIPS 203 Table 2 (ML-KEM-768); confirm at Phase 0)
ok      ETA1   = 2 (locked 2; FIPS 203 Table 2; confirm at Phase 0)
ok      ETA2   = 2 (locked 2; FIPS 203 Table 2; confirm at Phase 0)
ok      DU     = 10 (locked 10; FIPS 203 Table 2; confirm at Phase 0)
```

```

ok      DV      = 4  (locked 4; FIPS 203 Table 2; confirm at Phase 0)
ok      EK_BYTES = 1184 (locked 1184; verified: IETF draft-sfluhrer ... Table 1)
ok      DK_BYTES = 2400 (locked 2400; verified: same)
ok      CT_BYTES = 1088 (locked 1088; verified: same)
ok      SS_BYTES = 32  (locked 32; verified: same)

```

```

check_params: all locked parameters match.
exit=0

```

5. Artifact inventory

Path	Lines	sha256 (12)	Git
rtl/ntt/base_case_multiply.sv	40	ec12ac8be1bd	committed
rtl/ntt/butterfly.sv	57	36b18e465d4e	committed
rtl/ntt/modmul_reduce.sv	30	7788fb087210	committed
rtl/ntt/ntt_core.sv	235	43c558b8622f	committed
rtl/ntt/ntt_pkg.sv	44	e8ebff73f62	committed
rtl/ntt/poly_mem.sv	39	5c56edf9faf1	committed
rtl/ntt/twiddle_rom.sv	285	b960f8cd923c	committed
scripts/gen_twiddle_rom.py	79	19a96f48e0a3	committed
tb/ntt/confestest.py	7	449c29117b42	committed
tb/ntt/run_ntt_tests.py	80	eb38f944c343	committed
tb/ntt/test_base_case_multiply.py	57	91b8b03c2f6f	committed
tb/ntt/test_butterfly.py	62	432dec14f747	committed
tb/ntt/test_modmul.py	40	a903a06daac9	committed
tb/ntt/test_ntt_core.py	157	5ca577d1c742	committed
formal/phase01-ntt/ntt_core_formal_top.sv	51	bac97af81286	committed
formal/phase01-ntt/ntt_core_props.sv	59	465b5b46b881	committed
formal/phase01-ntt/ntt_core_safety.sby	36	c5c39f73eb86	committed
quartus/phase01_ntt_c0/phase01_ntt_c0.qpf	2	805f639fa11d	NEW (uncommitted)
quartus/phase01_ntt_c0/C0.qsf	29	fa66cca6b334	NEW (uncommitted)
quartus/phase01_ntt_c0/C0.sdc	4	9c6a085dfa75	NEW (uncommitted)
quartus/phase01_ntt_c0/report_critical_paths.tcl	67	ccde7d413a5c	NEW (uncommitted)
quartus/phase01_ntt_c0/extract_c0_timing_evidence.py	350	2318526310a0	NEW (uncommitted)
quartus/phase01_ntt_c0/segment_path.py	82	6e5f2367e00c	NEW (uncommitted)
docs/evidence/phase01-ntt-baseline/cocotb_regression_2026-09-29.txt	33	2885136332da	committed
docs/evidence/phase01-ntt-baseline/formal_ntt_core_safety_2026-09-29.txt	64	4cb84b1f733a	committed
docs/evidence/phase01-ntt-baseline/quartus_C0_timing_analysis_2026-09-29.json	2965	9838c55cd808	NEW (uncommitted)
docs/evidence/phase01-ntt-baseline/quartus_C0_timing_analysis_2026-09-29.md	229	47517ab17bbb	NEW (uncommitted)
docs/evidence/phase01-ntt-baseline/test_plan.md	61	860cee614306	committed
docs/evidence/quartus/C0-20260929.md	54	aceaaffdb619c	NEW (uncommitted)
docs/results/result_phase1.md	160	5237819ecd9d	MODIFIED (uncommitted)
docs/ROADMAP.md	401	d6e4e4f02b4d	MODIFIED (uncommitted)
docs/guides/quartus_phase1_compile_guide.md	-	-	DELETED in working tree

Quartus build products (output_files/, db/, incremental_db/, the .sof) are git-ignored and not listed; only the extracts in docs/evidence/ are the recorded evidence.

6. Decisions needed and next steps

- **Target clock ADR** (required at the Phase 1 gate): C0 cannot run at the provisional 50 MHz.
- **How C0 closes:** accept C0 as the documented failing baseline (fixes happen in Phases 2, 4, 5 as the roadmap orders them), or change C0 before freezing it. Either is a team decision; no RTL was changed.

- **Reading of CRG-9** ('or the failure documented'): FAIL (as recorded) or PASS-with-documented-failure.
- **Tooling**: set QUARTUS_BIN in scripts/tooling.env (currently empty; Quartus is at ~/altera_lite/25.1std).
- **Commit** the new Quartus project files, drill-down scripts and evidence (listed as NEW / MODIFIED above).
- **Check**: docs/guides/quartus_phase1_compile_guide.md is committed (d76c829) but deleted in the working tree; it was not restored by this report. Keep or restore it deliberately before committing.
- **Approval**: a team member ticks the box in docs/results/result_phase1.md after review.

7. Reproduce

```
. scripts/env.sh
verilator --lint-only -Wall --timing -sv rtl/ntt/*.sv --top-module ntt_core
slang --top ntt_core rtl/ntt/*.sv
python3 tb/ntt/run_ntt_tests.py icarus && python3 tb/ntt/run_ntt_tests.py verilator
python3 -m pytest tb/golden/tests/ -q && python3 .claude/skills/mlkem-guard/scripts/check_params.py
(cd formal/phase01-ntt && sby -f ntt_core_safety.sby)
export PATH=$HOME/altera_lite/25.1std/quartus/bin:$PATH
(cd quartus/phase01_ntt_c0 && quartus_sh --flow compile phase01_ntt_c0 -c C0)
(cd quartus/phase01_ntt_c0 && quartus_sta -t report_critical_paths.tcl && python3 extract_c0_timing_evidence.py)
python3 .claude/skills/phase-gate/scripts/check_result.py docs/results/result_phase1.md
```

Commits on this branch since Phase 0

```
3d49304 docs(fase1): test plan Fase 1 (CRG-4) sebelum RTL/tes ditulis
6fb61de rtl(fase1): baseline NTT/INTT config C0 -- L=1, satu butterfly per siklus
d29f90c tb(fase1): cocotb bit-exact vs golden model, 10/10 lulus di Icarus dan Verilator
160e964 formal(fase1): bukti keamanan FSM ntt_core via SymbiYosys k-induction (CRG-8)
d8cb3cd docs(fase1): log evidence regresi cocotb dan bukti formal
5d5ee74 docs(fase1): result_phase1.md (PARTIAL) + perbarui status ROADMAP
d76c829 docs: panduan compile Quartus manual untuk Fase 1 config C0
```